A COUNTEREXAMPLE, AS PRINTED, TO THE SINGH–SINGH WIDTH-2 ROABP MODULAR-STABILITY PROPOSITION

Abstract. Singh and Singh (arXiv:2602.13449v1) study the modular substitution $\Gamma_g\colon x_i \mapsto \lambda^{g^{\,i}}$ into $\mathcal{R}_r = K[\lambda]/\langle\lambda^r - 1\rangle$ and its bad set $B_r(C) = \{g \in \mathbb{Z}_r^* : \Gamma_g(C) = 0\}$. Their Conjecture 4.7.2 asserts $|B_r(C)| \le r^{1-\varepsilon}$ for every nonzero width-w, degree-d ROABP C and every prime $r \ge R(w,d)$, and their Proposition 4.7.3 asserts that conjecture, unconditionally, for width-2 diagonal ROABPs. For any prime $r \ne \operatorname{char} K$ and any $n \ge r$, the polynomial $C = x_1 - x_r$ is a nonzero width-2 diagonal ROABP of individual degree 1 with $B_r(C) = \mathbb{Z}_r^*$, so $|B_r(C)| = r-1$, the largest value possible; more generally $B_r(x_i - x_j) = \{g : g^{\,j-i} = 1\}$, of order $\gcd(j-i, r-1)$. The mechanism is Fermat's little theorem: $g^{\,r} \equiv g \pmod r$. Since Proposition 4.7.3 is unconditional, one witness in its class falsifies it as printed; letting r range over the primes falsifies the conjecture as it is quantified. The witness is a difference of two monomials, and thus lies in the degenerate class that the authors' own Remark on p. 27 removes from the conjecture's intended scope; the regime that Remark describes — C of width at least 3 with non-degenerate coefficient support — is untouched here and remains open.

1. The statements at issue, and the counterexample

Throughout, K is a field, r is a prime with $r \ne \operatorname{char} K$ (a condition that is automatic when $\operatorname{char} K = 0$), $\mathcal{R}_r = K[\lambda]/\langle\lambda^r - 1\rangle$, and $\mathbb{Z}_r^* = \{1, \dots, r-1\}$ is the multiplicative group modulo r. We follow [1, Definition 4.7.1, printed p. 26]: for $g \in \mathbb{Z}_r^*$ the *modular substitution* Γ_g is the K-algebra map

$$\Gamma_g\colon K[x_1, \dots, x_n] \longrightarrow \mathcal{R}_r, \qquad x_i \longmapsto \lambda^{g^{\,i}},$$

so that a monomial $\prod_i x_i^{a_i}$ is sent to λ^e with $e \equiv \sum_i a_i g^{\,i} \pmod r$. The exponent of g is the *index* i of the variable, nested; it is not the product $g \cdot i$. We write $C_g := \Gamma_g(C)$ and

$$B_r(C) := \{\, g \in \mathbb{Z}_r^* \;:\; C_g = 0 \text{ in } \mathcal{R}_r \,\}.$$

Since $\lambda^0, \lambda^1, \dots, \lambda^{r-1}$ is a K-basis of $\mathcal{R}_r$, deciding $C_g = 0$ is deciding whether the r coefficients obtained by collecting the monomials of C into residue classes of exponents modulo r all vanish.

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Definition 1. A *read-once oblivious algebraic branching program* (ROABP) of width w and individual degree d in the variable order $x_1, \dots, x_n$ computes $C = u^{\mathsf{T}} M_1(x_1) M_2(x_2) \cdots M_n(x_n)\, v$, where $u, v \in K^w$ and each M_i is a $w \times w$ matrix whose entries are univariate polynomials in x_i of degree at most d. It is *diagonal* if every M_i is a diagonal matrix; a width-2 diagonal ROABP therefore computes exactly a sum of two products of univariate polynomials.

The passages below are transcribed by hand from the printed pages of [1], which is PDF-only; only the mathematical symbols are reset in TeX. The source reuses the label 4.7.3 (an Example on p. 27, a Proposition on p. 28) and prints Remarks on both p. 27 and p. 28, so each reference names the page and the kind. [1, Conjecture 4.7.2, printed pp. 26–27], in full:

> **4.7.2 Conjecture (modular stability / hinge conjecture).** There exists an explicit polynomial $R(w,d)$ and an absolute constant $\varepsilon > 0$ such that for every nonzero polynomial C computed by a width-w, degree-d ROABP, and for every prime $r \geq R(w,d)$ (e.g., typically $r \geq (wd)^7$), the bad set $B_r(C) := \{g \in \mathbb{Z}_r^* : C_g \equiv 0 \text{ in } K[\lambda]/\langle \lambda^r - 1\rangle\}$ satisfies $|B_r(C)| \leq r^{1-\varepsilon}$.

As printed, the threshold carries no n: it depends on w and d only, so a witness with $n \geq r$ is admissible. One page later, [1, Proposition 4.7.3, printed p. 28] states, in full:

> **Proposition 4.7.3 (Special Case).** Conjecture 4.7.2 holds for width-2 diagonal ROABPs. *Proof:* In the diagonal case, coefficients are sums of scalars. Modular reduction acts as a hash on exponents, and standard sparse polynomial results apply.

The source also prints passages hedging the conjecture's scope; their bearing is taken up in Section 2.

Theorem 1. *Let K be a field, let r be a prime with $r \neq \operatorname{char} K$, and let $1 \leq i < j \leq n$. Then*

$$B_r(x_i - x_j) = \{\, g \in \mathbb{Z}_r^* \;:\; g^{\,j-i} = 1 \,\},$$

a subgroup of $\mathbb{Z}_r^*$ of order $\gcd(j-i,\, r-1)$. *In particular, if $n \geq r$ and $C = x_1 - x_r$, then $B_r(C) = \mathbb{Z}_r^*$ and $|B_r(C)| = r - 1$.*

Proof. Fix $g \in \mathbb{Z}_r^*$. By definition $C_g = \lambda^a - \lambda^b$ with $a \equiv g^{\,i}$ and $b \equiv g^{\,j} \pmod r$, the exponents reduced into $\{0, \dots, r-1\}$. Since $\lambda^0, \dots, \lambda^{r-1}$ is a K-basis of $\mathcal{R}_r$, we have $C_g = 0$ if and only if $a = b$, that is, if and only if $g^{\,i} \equiv g^{\,j} \pmod r$; this step uses only that the two coefficients 1 and -1 are nonzero in K. As g is invertible modulo r, $g^{\,i} \equiv g^{\,j}$ is equivalent to $g^{\,j-i} \equiv 1$. The set of such g is the kernel of the endomorphism $g \mapsto g^{\,j-i}$ of the cyclic group $\mathbb{Z}_r^*$ of order $r-1$, hence a subgroup, and its order is $\gcd(j-i, r-1)$.

For the last sentence take $i = 1$, $j = r$, so $j-i = r-1$ and $\gcd(r-1, r-1) = r-1$; equivalently, $g^r \equiv g \pmod r$ for every g by Fermat's little theorem, so every $g \in \mathbb{Z}_r^*$ is bad. $\square$

Proposition 2. *For any $1 \le i < j \le n$ the polynomial $C = x_i - x_j$ is nonzero over every field, has exactly two monomials, and is computed by a width-*2 diagonal *ROABP of individual degree* 1 *in the variable order $x_1, \ldots, x_n$.*

Proof. Put $u = (1,1)^\mathsf{T}$, $v = (1,-1)^\mathsf{T}$ and, for $1 \le t \le n$,

$$M_t(x_t) = \operatorname{diag}(\alpha_t, \beta_t), \qquad \alpha_t = \begin{cases} x_t, & t = i \\ 1, & t \ne i \end{cases} \qquad \beta_t = \begin{cases} x_t, & t = j \\ 1, & t \ne j \end{cases}$$

Every M_t is diagonal with entries of degree at most 1 in x_t, and $\prod_t M_t = \operatorname{diag}(x_i, x_j)$, so $u^\mathsf{T}(\prod_t M_t)v = x_i - x_j$. Nonzeroness is immediate: x_i and x_j are distinct monomials and their coefficients $1, -1$ are nonzero in every field. In characteristic 2 the polynomial is $x_i + x_j$, which is the same object. $\square$

Corollary 3 (Proposition 4.7.3 is false as printed)**.** *Let r be a prime with $r \ne \operatorname{char} K$, let $n \ge r$ and let $C = x_1 - x_r$. Then C is a nonzero width-*2 *diagonal ROABP of individual degree* 1 *with $|B_r(C)| = r-1$. Consequently, for a given $\varepsilon > 0$, the bound of* [1, Conjecture 4.7.2] *fails on C at every such r with $r - 1 > r^{1-\varepsilon}$, and in particular at every $r > 2^{1/\varepsilon}$. Hence* [1, Proposition 4.7.3, printed p. 28] *is false as printed.*

Proof. Theorem 1 gives $|B_r(C)| = r-1$ and Proposition 2 places C in the class. For the last inequality, $r - 1 \ge r/2$ for $r \ge 2$, and $r/2 > r^{1-\varepsilon}$ is $r^\varepsilon > 2$. $\square$

Corollary 4 (the printed form of Conjecture 4.7.2 is false)**.** *For every candidate pair (R, ε) with $\varepsilon > 0$ there are a nonzero width-*2*, degree-*1 *ROABP C and a prime $r \ge R(2,1)$ with $|B_r(C)| > r^{1-\varepsilon}$. It suffices to pick any prime r with $r \ge R(2,1)$, $r > 2^{1/\varepsilon}$ and $r \ne \operatorname{char} K$, then $n \ge r$ and $C = x_1 - x_r$, and apply Corollary 3. Hence* [1, Conjecture 4.7.2] *is false as printed, that is, as a statement quantified over all nonzero width-w, degree-d ROABPs and all primes $r \ge R(w,d)$.*

2. Scope

The source itself puts objects of this shape outside the conjecture's intended scope. Its p. 27 Remark observes that differences of two monomials can be annihilated by Γ_g, calls such constructions "highly structured and algebraically degenerate" and of "extremely low algebraic density", and says that Conjecture 4.7.2 "concerns the complementary regime"; the p. 28 Remark repeats that reading. Now $C = x_1 - x_r$ has width 2 and support size 2, so it lies in the class those passages exclude. Against the statement they describe — C of width at least 3 with non-degenerate coefficient support — nothing here bears, and that statement remains open; we offer no evidence either way about it. What those passages do not do is qualify Proposition 4.7.3, which carries no degeneracy hypothesis, so one witness in its class suffices. No repair of Proposition 4.7.3 is offered here.

The witness object is not new. [1, Example 4.7.3, printed p. 27] already states that if $S \neq S'$ are subsets of $[n]$ with $\sum_{i\in S} g^i \equiv \sum_{i\in S'} g^i \pmod r$, then $C = \prod_{i\in S} x_i - \prod_{i\in S'} x_i$ satisfies $C_g = 0$; Theorem 1 is that Example at $S = \{1\}$, $S' = \{r\}$, where the hypothesis reads $g \equiv g^r \pmod r$ and so holds for every g simultaneously. Likewise $B_r(x_i - x_j)$ is the kernel of $g \mapsto g^{j-i}$ on the cyclic group $\mathbb{Z}_r^*$, and the order of such a kernel is $\gcd(j-i, r-1)$: this is textbook. We claim novelty neither for the object nor for the count.

3. Verification

Theorem 1 is a two-line argument in the cyclic group $\mathbb{Z}_r^*$ and needs no machine, so the accompanying program `verify.py` is corroboration at particular parameters and not a premise. It implements Γ_g directly from Definition 4.7.1, building C_g as a full length-r coefficient vector over the basis $\lambda^0, \ldots, \lambda^{r-1}$; it obtains $B_r(x_1 - x_r) = \mathbb{Z}_r^*$ at $r = 31, 131, 1039$ over characteristics $0, 2, 3$ and, by the exponent-congruence route, also at $r = 4099$ and $r = 10007$; it confirms the subgroup order $\gcd(j-i, r-1)$ for every index difference at those primes; and it evaluates the explicit width-2 diagonal matrix product of Proposition 2 symbolically. It uses only the Python standard library and exact integer arithmetic.

The program covers finitely many primes at the single value $\varepsilon = 1/10$ and the source's illustrative threshold $R(2,1) = 128$; the quantification over all $\varepsilon > 0$ and all candidate thresholds in Corollary 4 rests on the proof of Theorem 1 alone, not on the run. Its output also records what it does not cover: no ROABP of width at least 3 and no object of non-degenerate coefficient support is tested, no sweep at $n \ll r$ is offered, and the source PDF is not fetched, so the passages quoted above are hand transcriptions from the printed pages named — no program here can re-check a transcription, and a reader who doubts one should compare it against the PDF by eye.

References

[1] S. Singh and V. Singh, *An Algebraic Rigidity Framework for Order-Oblivious Deterministic Black-Box PIT of ROABPs*, arXiv:2602.13449v1, 13 February 2026, 48 pp. arXiv:2602.13449. The submission is PDF-only; page references are to the printed pages of that PDF.